



Hypothyroidism

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Hypothyroidism is underactivity of the thyroid gland that leads to inadequate production of thyroid hormones and a slowing of vital body functions.

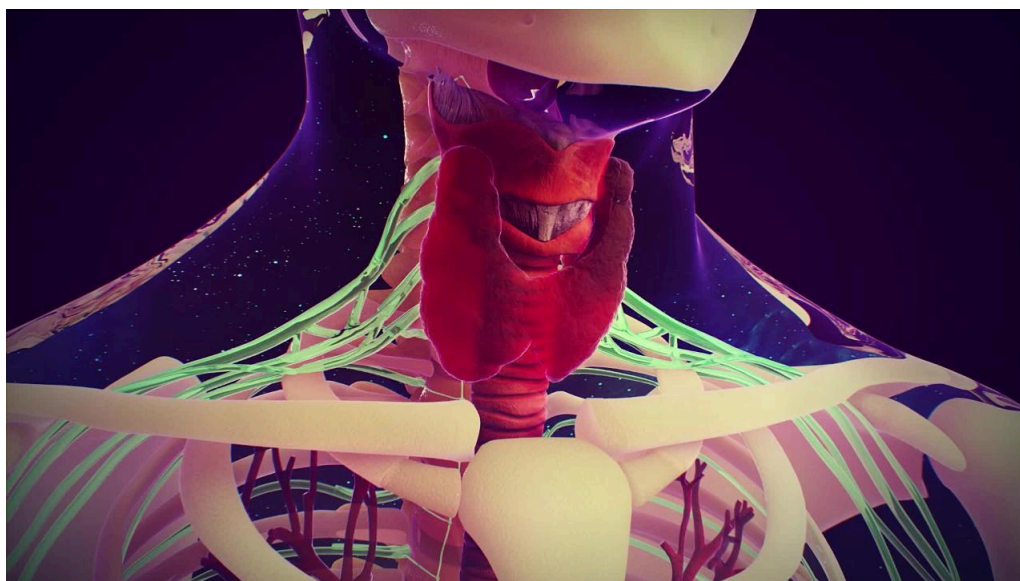
- Facial expressions become dull, the voice is hoarse, speech is slow, eyelids droop, and the eyes and face become puffy.
- Usually only one blood test is needed to confirm the diagnosis.
- Most people with hypothyroidism need to take thyroid hormone for the rest of their life.

The thyroid gland secretes thyroid hormones, which control the speed at which the body's chemical functions proceed (metabolic rate). Thyroid hormones affect many vital body functions, such as the heart rate, the rate at which calories are burned, skin maintenance, growth, heat production, fertility, and digestion. There are 2 thyroid hormones:

- T4: Thyroxine (also called tetraiodothyronine)
- T3: Triiodothyronine

The Thyroid

VIDEO



The pituitary gland produces thyroid-stimulating hormone (TSH), which stimulates the thyroid gland to produce thyroid hormones. The pituitary gland slows or speeds the release of TSH, depending on whether the levels of thyroid hormones circulating in the blood are getting too high or too low. (See also [Overview of the Thyroid Gland](#).)

Hypothyroidism is common, especially among older adults, particularly women. It affects about 10% of older women. It can, however, occur at any age.

Myxedema is the name given to very severe hypothyroidism.

Causes of Hypothyroidism

Hypothyroidism may be:

- Primary
- Secondary

Primary hypothyroidism results from a disorder of the thyroid gland itself. In the United States, the **most common** cause is:

- [Hashimoto thyroiditis](#): Hypothyroidism develops as the thyroid gland is gradually destroyed.

Other causes of primary hypothyroidism include:

- Thyroid inflammation (thyroiditis other than Hashimoto thyroiditis)
- Treatment of [hyperthyroidism](#) or [thyroid cancer](#)
- Lack of iodine
- Radiation to the head and neck
- Inherited disorders that prevent the thyroid gland from making or secreting enough hormones

Thyroid inflammation (thyroiditis) can cause permanent or temporary hypothyroidism depending upon the cause. Hashimoto thyroiditis gradually destroys the thyroid gland and usually causes permanent hypothyroidism. [Subacute thyroiditis](#) is probably caused by an infection with a virus. Autoimmune inflammation that occurs after childbirth ([silent lymphocytic thyroiditis](#)) is another cause. In subacute thyroiditis and silent lymphocytic thyroiditis, the hypothyroidism is usually temporary because the thyroid gland is not destroyed.

Treatment of [hyperthyroidism](#) or [thyroid cancer](#) can cause hypothyroidism because radioactive iodine or medications used in treatment interfere with the body's ability to make thyroid hormones. Surgical removal of the thyroid gland leads to a lack of thyroid hormone production.

A **chronic lack of iodine** in the diet is the most common cause of hypothyroidism in countries where iodine is not added to salt. However, [iodine deficiency](#) is a rare cause of hypothyroidism in the United States and some other countries because iodine is added to table salt and is also used to sterilize the udders of dairy cattle and thus is present in dairy products.

Radiation to the head and neck, usually given as radiation therapy to treat cancer, can also cause hypothyroidism.

Rarer causes of hypothyroidism include some inherited disorders in which an abnormality of the enzymes in thyroid cells prevents the gland from making or secreting enough thyroid hormones (see also [Hypothyroidism in Infants and Children](#)).

Secondary hypothyroidism occurs when the pituitary gland fails to secrete enough thyroid-stimulating hormone (TSH), which is necessary for normal stimulation of the thyroid. Secondary hypothyroidism is much rarer than primary.

Symptoms of Hypothyroidism

Insufficient thyroid hormones cause body functions to slow. Symptoms are subtle and develop gradually. Some of them may be mistaken for depression, especially among older adults.

- Facial expressions become dull.
- The voice is hoarse, and speech is slow.
- Eyelids droop.
- The eyes and face become puffy.
- The hair becomes sparse, coarse, and dry.
- The skin becomes coarse, dry, scaly, and thick.

Many people with hypothyroidism are fatigued, gain weight, become constipated, develop muscle cramps, and are unable to tolerate cold. Some people develop [carpal tunnel syndrome](#), which makes the hands tingle or hurt. The pulse may slow, the palms and soles may appear slightly orange (carotenemia), and the side parts of the eyebrows slowly fall out. Some people, especially older adults, may appear confused or forgetful—signs that can easily be mistaken for [Alzheimer disease](#) or other forms of dementia. They may appear depressed. Women with hypothyroidism may have changes in their menstrual periods.

Facial Changes in Hypothyroidism

IMAGE



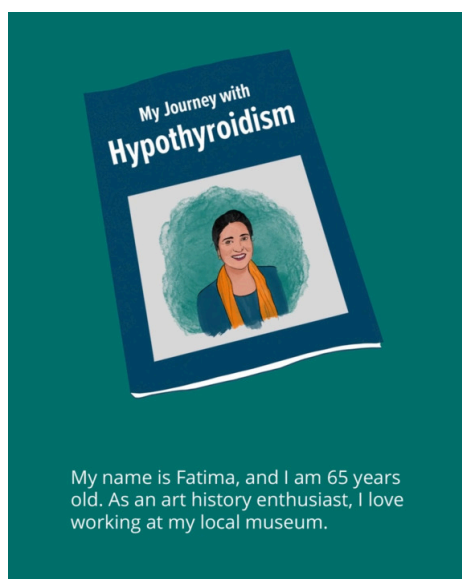
In people with hypothyroidism, the facial expression becomes dull, the eyelids droop, and the eyes and face become puffy.

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People with hypothyroidism frequently have [high levels of cholesterol](#) in their blood.

Hidden Disability: Hypothyroid...

VIDEO



Myxedema coma

If untreated, hypothyroidism can eventually cause [anemia](#), a low body temperature, and [heart failure](#). This situation may progress to confusion, stupor, or coma (myxedema coma). Myxedema coma is a life-threatening complication in which breathing slows, seizures occur, and blood flow to the brain decreases. Myxedema coma can be triggered in a person with hypothyroidism by physical stresses, such as exposure to the cold, as well as by an infection, injury, surgery, and medications such as sedatives that depress brain function.

Spotlight on Aging: Hypothyroidism in Older Adults

Many older adults have some degree of hypothyroidism. About 10% of women and 6% of men are affected.

Typical symptoms of hypothyroidism, such as weight gain, muscle cramps, tingling of the hands, and the inability to tolerate cold, are less common among older adults. When such symptoms do occur among older adults, they are sometimes less obvious.

Older adults may also have less typical symptoms. For example, they may lose weight or become confused, and they may have a decreased appetite, joint stiffness, joint or muscle pains, weakness, and a tendency to fall.

Because symptoms in older adults can be different from those in younger adults, are often subtle and vague, and are common among older adults who do not have hypothyroidism, doctors may not recognize these symptoms as being caused by hypothyroidism. Some experts recommend screening for thyroid disease every year or every few years in people older than 70 years. However, several medical organizations have examined the pros and cons of such screening and recommend *against* such screening in people without any suggestive symptoms or signs of thyroid disease in order to avoid over-treating people with minor laboratory abnormalities.

Diagnosis of Hypothyroidism

- Measurement of thyroid-stimulating hormone (TSH) levels in blood

Doctors usually suspect hypothyroidism on the basis of the symptoms and findings on physical examination, including a slow pulse.

LAB TEST

[TSH \(Thyroid-Stimulating Hormone\) Test](#)



Usually hypothyroidism can be diagnosed with one simple blood test: the measurement of TSH. If the thyroid gland is underactive, the level of TSH is high.

In those rare cases of hypothyroidism caused by inadequate secretion of TSH, a second blood test is needed. This blood test measures the level of the thyroid hormone T4 (thyroxine, or tetraiodothyronine). A low level supports the diagnosis of hypothyroidism if the free level of T4 is also low. In that case, evaluation of the pituitary with pituitary function tests and imaging is generally performed.

Treatment of Hypothyroidism

- Replacement of thyroid hormone

Treatment involves replacing thyroid hormone using one of several oral preparations. The preferred form of hormone replacement is synthetic T4 (levothyroxine). Another form, desiccated (dried) thyroid, is obtained from the thyroid glands of animals but is no longer used very often. In general, desiccated thyroid is less satisfactory than synthetic T4 because the content of thyroid hormones in the tablets may vary.

In emergencies, such as myxedema coma, doctors may give synthetic T4, T3 (triiodothyronine), or both intravenously.

Treatment begins with small doses of thyroid hormone, because too large a dose can cause serious side effects, although large doses may eventually be necessary. The starting dose and the rate of increase are especially small in older adults, who are often most at risk of side effects. The dose is gradually increased until the levels of TSH in the person's blood return to normal.

In patients being treated for hypothyroidism, doses usually need to be increased during pregnancy.



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